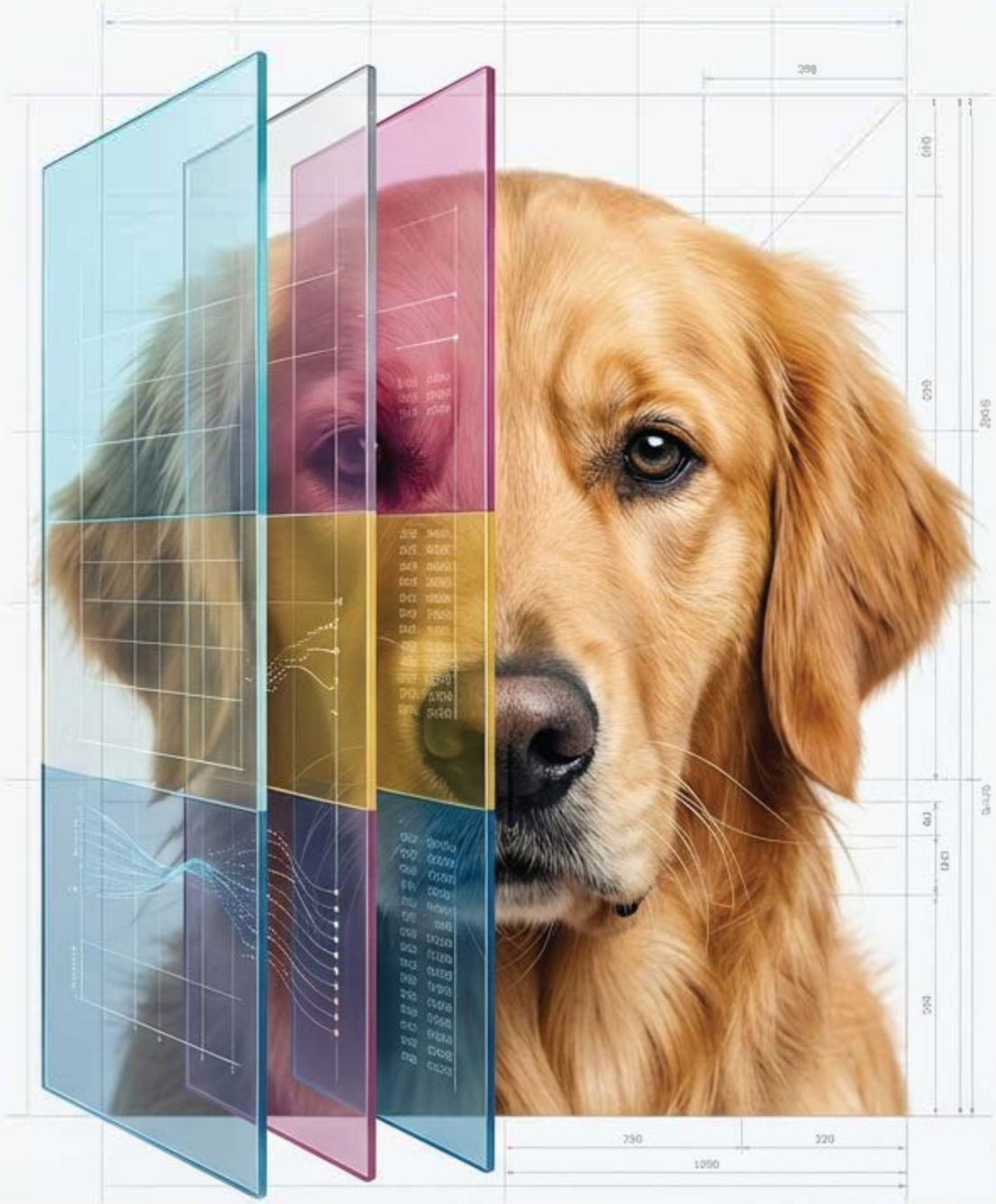
High-Fidelity Dog Breed Classification Without End-to-End Training

Achieving 94.8% accuracy across 120 breeds using multi-backbone CNN feature fusion, feture fusion,PCA, and a Linear Support Vector Machine.



Bypassing Expensive Neural Network Training Through Strategic Fusion

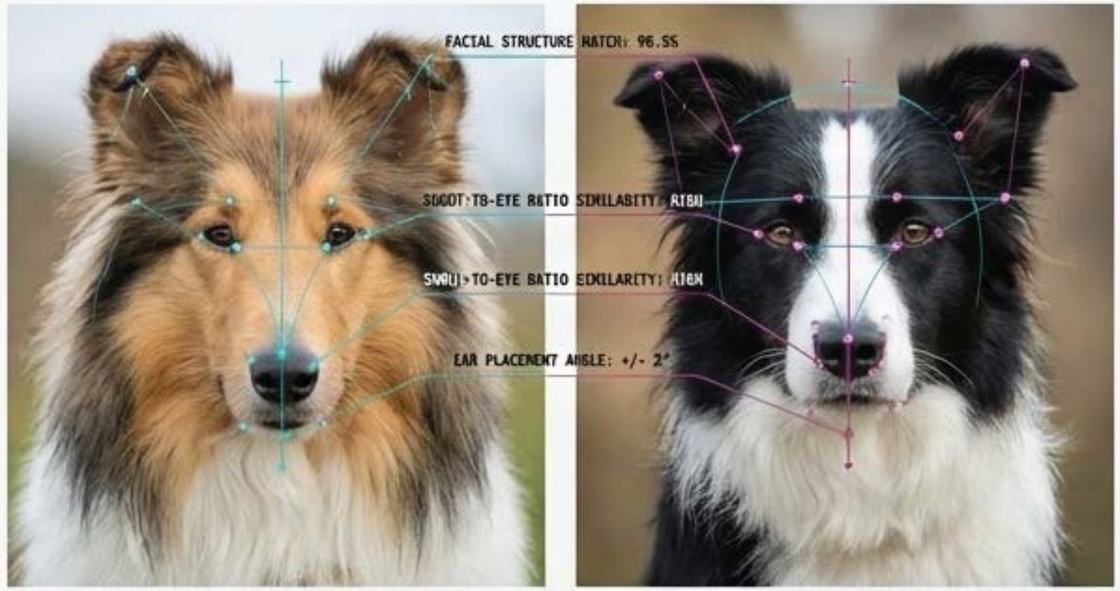
Classifying highly similar dogs from a dataset of 20,580 images.

Extracting visual knowledge from models already trained on 1.2M ImageNet images.

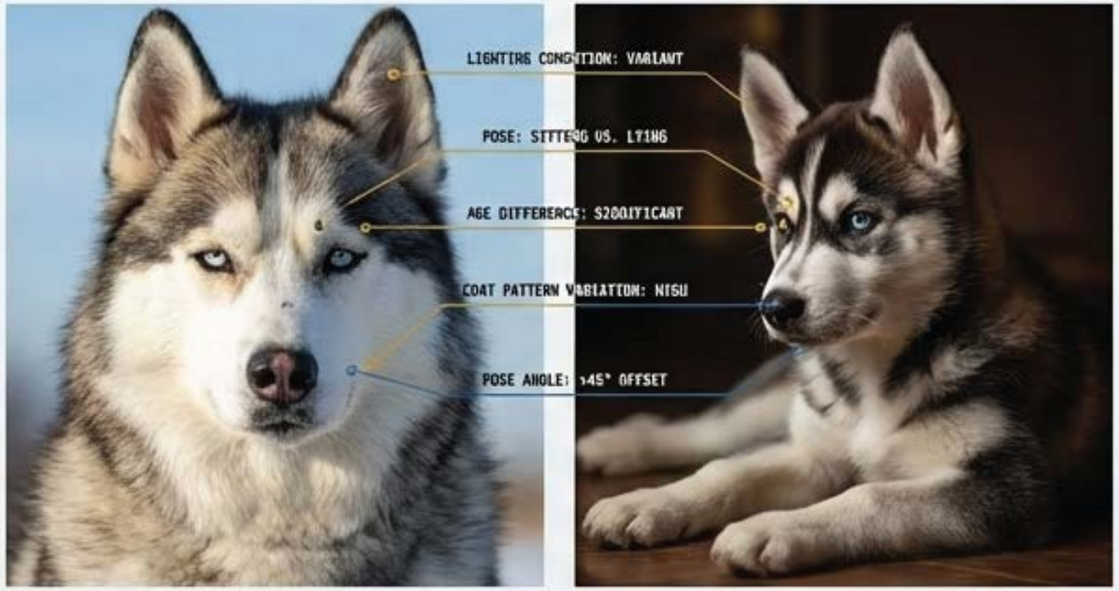
Achieved using classical machine learning on top of merged neural network outputs.

Fine-Grained Recognition Demands High Nuance With Limited Data

Inter-Class Similarity

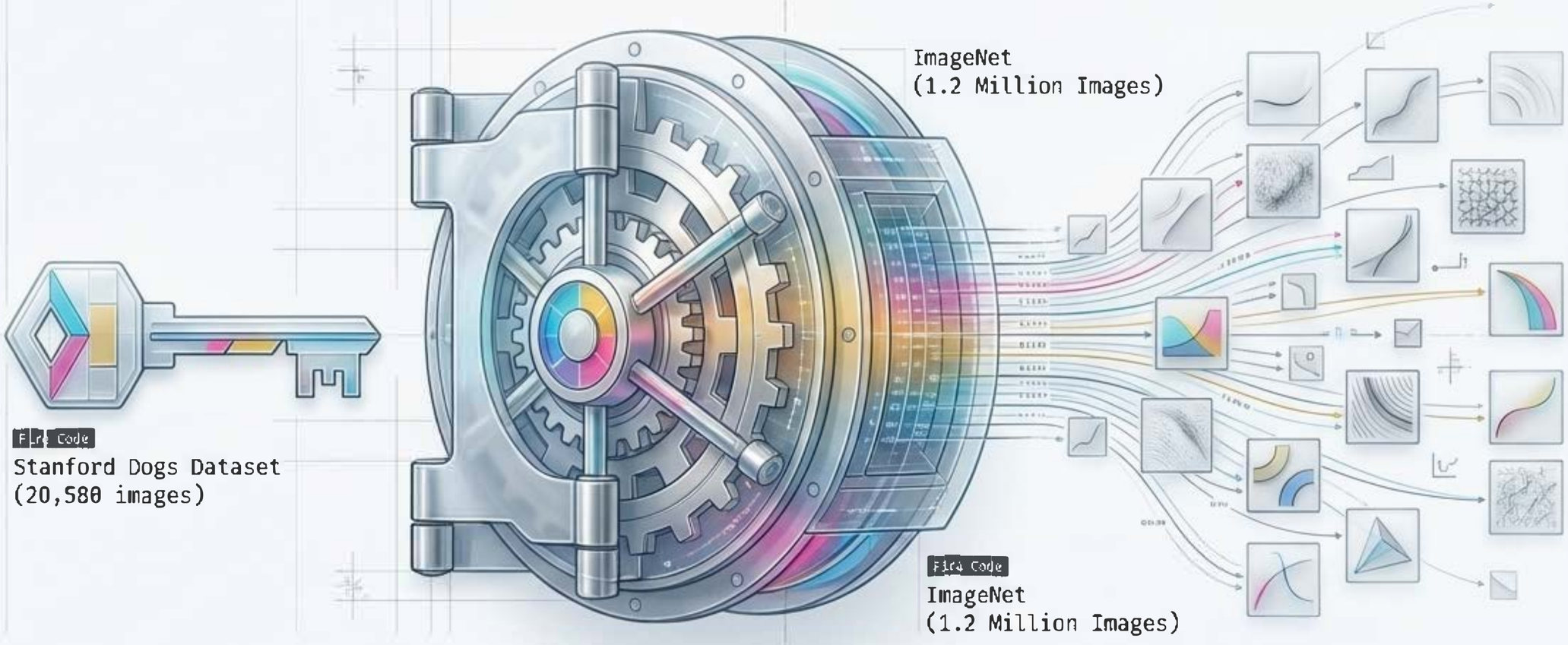


Intra-Class Variance



The primary constraint: Building a system to detect these subtle differences using only ~150 training images per breed without severe overfitting.

Leveraging Pre-Learned Visual Vocabularies



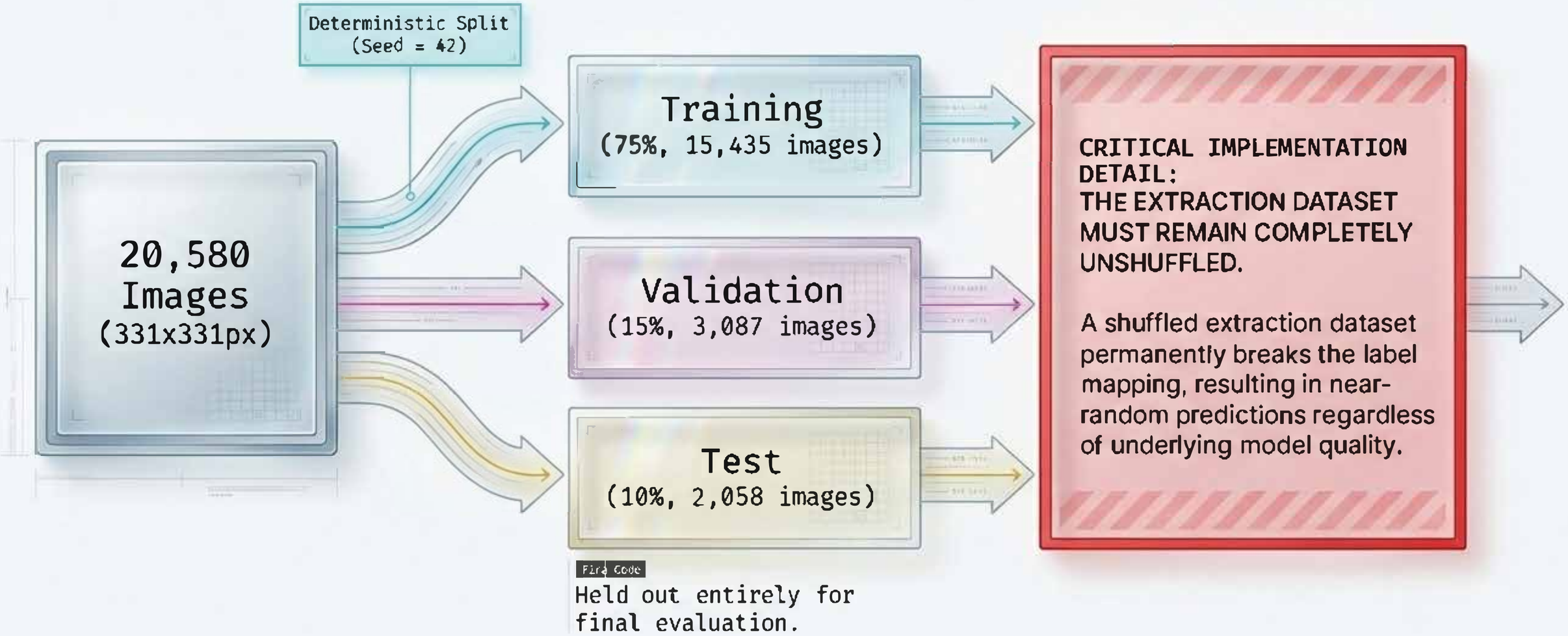
Rather than training a neural network from scratch on small data, we utilize the extracted visual vocabulary of state-of-the-art deep learning models as frozen feature extractors.

Four Distinct CNN Architectures Act as Complementary Lenses

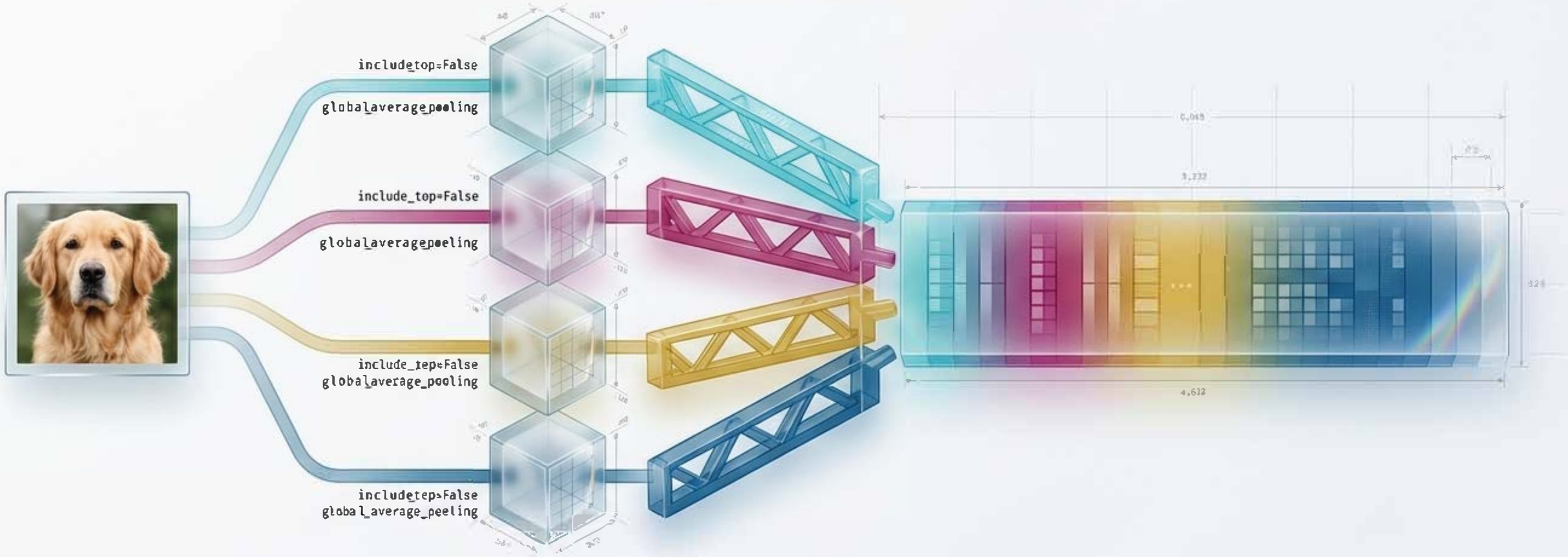
InceptionV3	Vector Size: 2,048	Architecture Style: Factored convolutions
InceptionResNetV2	Vector Size: 1,536	Architecture Style: Inception + residual connections
NASNetLarge	Vector Size: 4,032	Architecture Style: Neural Architecture Search
EfficientNetB3	Vector Size: 1,536	Architecture Style: Compound scaled CNN

No single network captures every detail. Together, they provide a multi-dimensional description far richer than any solitary model.

Deterministic Data Splitting and the Unshuffled Extraction Requirement



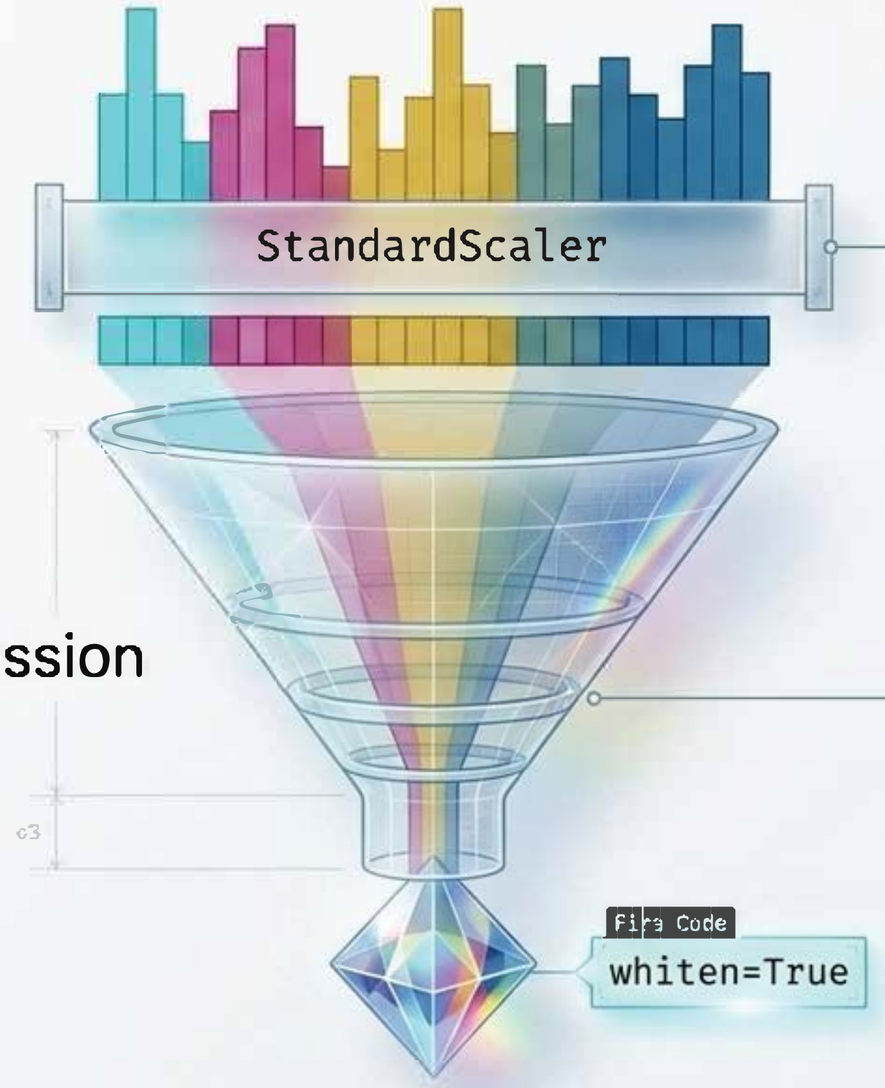
Fusing Four Neural Perspectives into a Single Descriptor



Process: Images are passed through frozen backbones utilizing global average pooling.
Output: 2,048 + 1,536 + 4,032 + 1,536 = A combined representation of 9,152 dimensions per image.

Compressing Features to 256 Dimensions via Standardization and PCA

Strict AI gate



PCA Compression

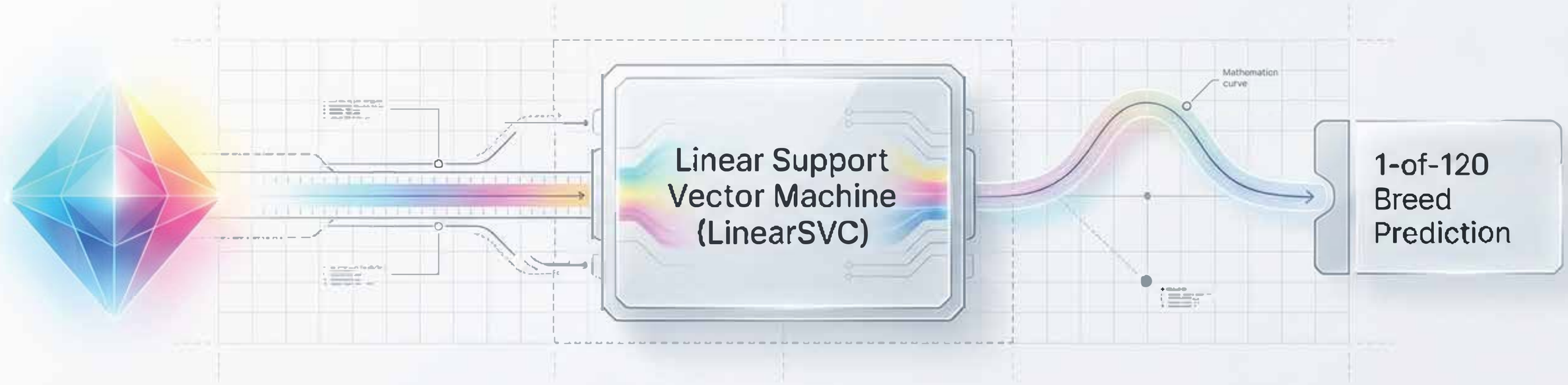
Fira Code

Transforms all features to zero mean and unit variance. Essential to prevent the backbone with the largest numerical activations from dominating.

Fira Code

Reduces the 9,152 dimensions down to 256 principal components, removing redundancy and producing a decorrelated feature space. (Fitted on training data only).

Rapid Classification Using a **Linear Support Vector Machine**



Algorithm:

Linear Support Vector Machine (LinearSVC).

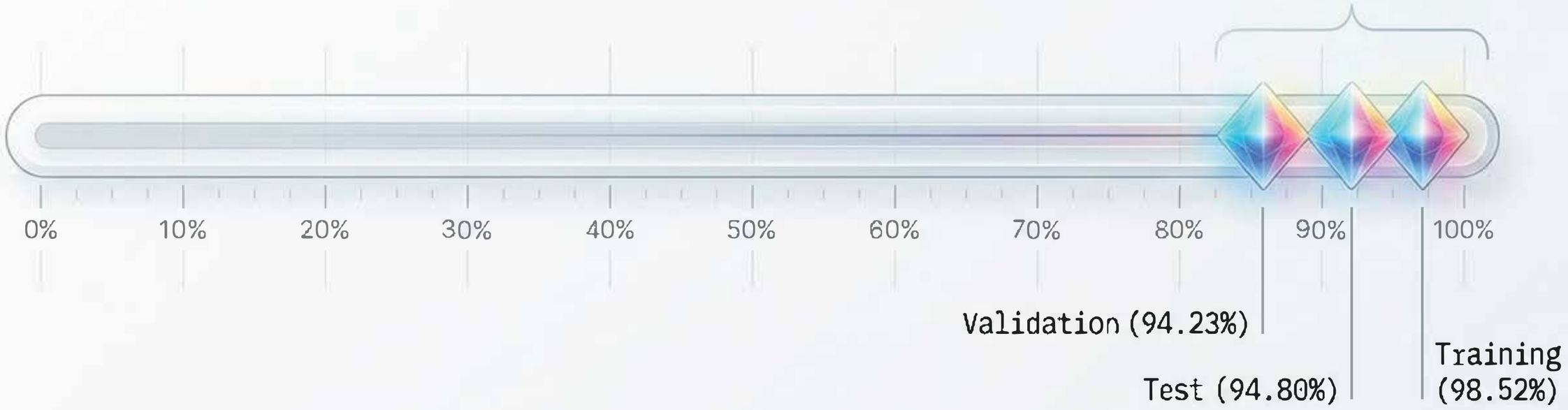
Hyperparameters:

Regularization parameter C=0.1.

Efficiency:

Because training samples (15,435) far exceed features (256), the dual=False primal formulation is utilized. The model achieves fast, successful convergence within 10,000 iterations.

Minimal Overfitting Demonstrated Across Validation Splits



Interpretation: The modest gap between 98.5% training and 94.8% test accuracy indicates robust generalization. The slight drop reflects the genuine, inherent difficulty of fine-grained breed discrimination, not severe overfitting.

Perfect Precision Contrasts With the Arctic Spitz Ambiguity

Flawless Recall (100%)

Toy terrier, Blenheim spaniel, Bluetick, Norwegian elkhound, Saluki, Airedale, Scotch terrier.

The Hardest Cases

The Bottom 10 F1-Scores

Eskimo dog (0.583, n=11)	Yorkshire terrier (0.800, n=7)
Siberian husky (0.682, n=24)	Shih-Tzu (0.800, n=16)
Lhasa (0.759, n=16)	Malamute (0.813, n=16)
Collie (0.769, n=15)	Walker hound (0.815, n=13)
Yorkshire terrier (0.800, n=7)	Am. Staffordshire terrier (0.833, n=17)
Shih-Tzu (0.800, n=16)	Border collie (0.846, n=11)

Global Average Pooling Creates a Blind Spot for Spatial Nuance

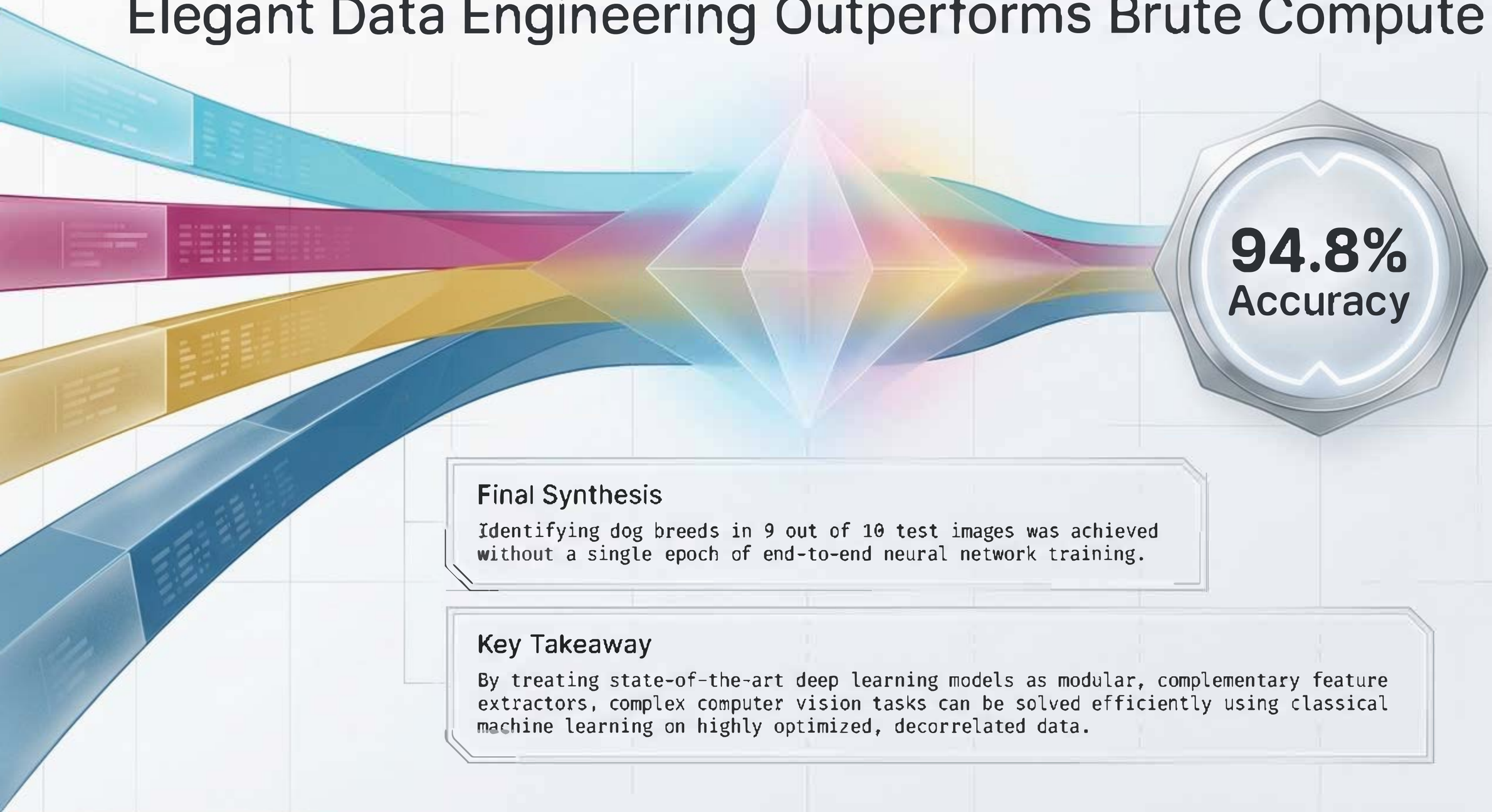
The Diagnostic Insight

The pipeline fails primarily on the Arctic Spitz family (Husky, Malamute, Eskimo) and visually similar pairs (Lhasa Apso / Shih-Tzu).

The Root Cause

Global Average Pooling compresses the entire image into a single vector, explicitly discarding the precise spatial map of distinguishing micro-features (like ear shape and facial markings).

Elegant Data Engineering Outperforms Brute Compute



Final Synthesis

Identifying dog breeds in 9 out of 10 test images was achieved without a single epoch of end-to-end neural network training.

Key Takeaway

By treating state-of-the-art deep learning models as modular, complementary feature extractors, complex computer vision tasks can be solved efficiently using classical machine learning on highly optimized, decorrelated data.